9

Eigenvalues and Eigenvectors

Historical Note

Eigenvalues arrived through mechanics and astronomy, not algebra. Euler in the 1750s, studying how a rigid body spins, found that every body has three perpendicular axes about which it rotates without wobbling; those are the eigenvectors of its inertia tensor. Lagrange and Laplace then met the same computation in the theory of planetary perturbations, where the slow drifts of the orbits were governed by the roots of what they called the *secular* equation, from *saeculum*, a century, because the effects take that long to show. The name survives in our *characteristic* equation. Cauchy in 1829 proved that for a symmetric array those roots are always real, which is the theorem Part V will call the spectral theorem. The word *eigenvalue* is much later: it is Hilbert's *Eigenwert* of 1904, a German compound meaning own-value that English absorbed rather than translated, displacing Sylvester's *latent root* and the older *proper value*. One reason the subject took two centuries to settle is that its natural home is the complex numbers, and the eighteenth century was not comfortable there.

Chapter 7 argued that every question of the form "how simple can this matrix be made?" is a question about choosing a basis, and Example 7.25 showed a reflection turning from a matrix of fractions into $\operatorname{diag}(1,-1)$ once the basis was adapted to the geometry. That example worked because two particular directions were merely stretched by the map rather than turned. This chapter asks which maps have such directions, how to find them, and how many there can be.

9.1 Directions that are only stretched

Definition 9.1 (Eigenvalue, eigenvector, eigenspace)**.** *Let $T : V \to V$ be an operator. A scalar λ is an* eigenvalue *of T if there exists $v \neq \mathbf{0}$ with*

$$Tv = \lambda v,$$

and any such v is an eigenvector *for λ. The set*

$$E_\lambda = \ker(T - \lambda \,\mathrm{id}) = \{v \in V : Tv = \lambda v\}$$

is the eigenspace *of λ, a subspace by Proposition 7.9, and its dimension is the* geometric multiplicity *of λ. The set of all eigenvalues is the* spectrum*, written* $\mathrm{Spec}(T)$.

Figure 9.1 is the definition in a picture: an eigenvector is a direction the matrix does not turn, and almost every direction is turned. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ the vector $(1, 1)$ is sent three times along itself, while $(1, 0)$ comes back pointing somewhere else.

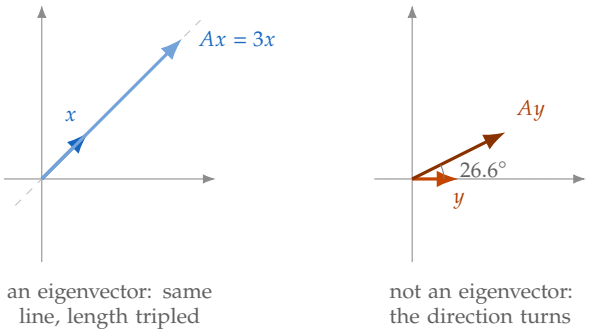


Figure 9.1. An eigenvector is a direction the matrix does not turn. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ the vector $x = (1, 1)$ is sent to $(3, 3)$, three times itself and along the same line; the vector $y = (1, 0)$ is sent to $(2, 1)$, which points 26.6 degrees away from where it started. Almost every direction turns, and the eigenvectors are the exceptional ones that do not. That is the whole definition, and the reason it matters is that along such a direction the matrix acts as a single number: applying A a hundred times to x multiplies it by 3^{100}, a computation with no matrices in it at all.

Caution

Two conventions in that definition are doing real work. The eigenvector must be nonzero: if $\mathbf{0}$ were allowed then $T\mathbf{0} = \lambda\mathbf{0}$ for every λ and the notion would be vacuous. But the eigen*value* may perfectly well be zero, and $0 \in \mathrm{Spec}(T)$ says exactly that $\ker T \neq \{\mathbf{0}\}$, that is, that T is not injective. The eigenspace E_λ does contain $\mathbf{0}$, since it is a subspace; it is the eigenvectors together with $\mathbf{0}$.

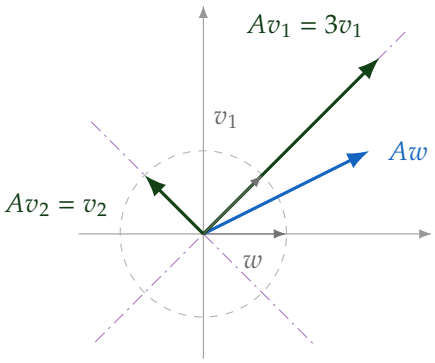


Figure 9.2. The action of $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ on three unit vectors, shown as grey arrows, with their images in colour. A generic direction such as w is turned. The two dash-dotted lines are not turned at all: along $y = x$ every vector is stretched by 3, and along $y = -x$ every vector is left exactly where it was, so the input and output arrows coincide and only one is drawn. Those two lines are the eigenspaces.

Example 9.2 (Eigenvalues by inspection). *(i) A diagonal matrix* $\operatorname{diag}(d_1, \dots, d_n)$ *has eigenvalues* $d_1, \dots, d_n$ *with eigenvectors* $e_1, \dots, e_n$.

(ii) The reflection of Example 7.25 has $Tu = u$ *and* $Tw = -w$, *so its eigenvalues are* 1 *and* -1 *with eigenspaces the mirror line and its perpendicular.*

(iii) The orthogonal projection onto a line L *fixes* L *and kills* $L^{\perp}$, *so its eigenvalues are* 1 *and* 0.

(iv) Rotation of the plane through an angle θ *with* $0 < \theta < \pi$ *has* no *real eigenvalue: it turns every direction, which is what rotating means.*

(v) The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ *fixes* e_1 *and turns everything else, so its only eigenvalue is* 1 *and its eigenspace is the single line* $\operatorname{span}\{e_1\}$.

(vi) On the space of smooth functions, differentiation satisfies $De^{at} = a\,e^{at}$, *so* every *real number is an eigenvalue of* D. *Infinite dimensions are a different world.*

9.2 Finding them: the characteristic polynomial

Proposition 9.3 (The eigenvalue condition). *For* $A \in M_n(\mathbb{R})$ *and a scalar* λ, *the following are equivalent:* λ *is an eigenvalue of* A; *the matrix* $A - \lambda\mathbf{1}$ *is not invertible;* $\det(A - \lambda\mathbf{1}) = 0$.

Proof. λ is an eigenvalue exactly when $(A - \lambda\mathbf{1})v = \mathbf{0}$ has a nonzero solution, that is, when $N(A - \lambda\mathbf{1}) \neq \{\mathbf{0}\}$. By Theorem 4.16 that is exactly the failure of invertibility, and

by Corollary 8.7 that is exactly $\det(A - \lambda \mathbf{1}) = 0$. □

Definition 9.4 (Characteristic polynomial). $p_A(\lambda) = \det(A - \lambda \mathbf{1})$.

Proposition 9.5 (Its shape). *p_A is a polynomial of degree exactly n, and*

$$p_A(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} (\operatorname{trace} A)\, \lambda^{n-1} + \cdots + \det A.$$

Proof. Apply the Leibniz formula of Theorem 8.10 to $A - \lambda \mathbf{1}$. The term of the identity permutation is $\prod_i (a_{ii} - \lambda)$, which expands to $(-1)^n \lambda^n + (-1)^{n-1} \big(\sum_i a_{ii}\big) \lambda^{n-1}$ plus lower-order terms. Any other permutation moves at least two indices, so its term omits at least two diagonal factors and has degree at most $n-2$. The constant term is $p_A(0) = \det A$. □

Key Idea

Some books write $\det(\lambda \mathbf{1} - A)$ instead, which is monic and differs from ours by the factor $(-1)^n$. Nothing depends on the choice, since the roots are the same, and the roots are the whole point. We keep $\det(A - \lambda \mathbf{1})$ because it makes $p_A(0) = \det A$ come out without a sign.

Proposition 9.6 (The characteristic polynomial belongs to the operator). *Similar matrices have the same characteristic polynomial. Hence an operator T on a finite-dimensional space has a well-defined p_T, computed from its matrix in any basis.*

Proof. $\det(P^{-1} A P - \lambda \mathbf{1}) = \det\big(P^{-1}(A - \lambda \mathbf{1}) P\big) = \det(A - \lambda \mathbf{1})$ by Proposition 8.13(c), using $P^{-1}(\lambda \mathbf{1}) P = \lambda \mathbf{1}$. □

Example 9.7 (A 2×2, in full). *For* $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$,

$$p_A(\lambda) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

So $\operatorname{Spec}(A) = \{1, 3\}$. *For* $\lambda = 3$, *solve* $(A - 3\mathbf{1})v = \mathbf{0}$ *with* $A - 3\mathbf{1} = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, *giving* $E_3 = \operatorname{span}\{(1, 1)\}$. *For* $\lambda = 1$, $A - \mathbf{1} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ *gives* $E_1 = \operatorname{span}\{(1, -1)\}$. *This is Figure 9.2. Check the two invariants:* $\operatorname{trace} A = 4 = 3 + 1$ *and* $\det A = 3 = 3 \cdot 1$, *as Theorem 9.12 will explain.*

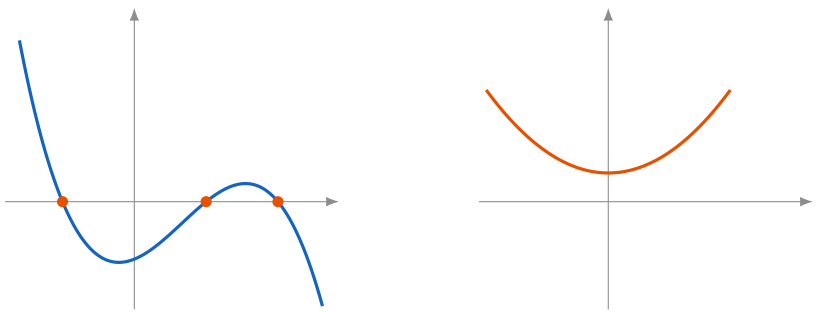
Example 9.8 (A 3×3 with a repeated eigenvalue)**.** *Let* $A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$. *Rather than expand a determinant, notice* $A = 2\mathbf{1} + J$ *where* J *is the all-ones matrix. Now* $J(1,1,1) = 3(1,1,1)$*, and* $Jx = \mathbf{0}$ *for every* x *with* $x_1 + x_2 + x_3 = 0$*, a two-dimensional space. So* J *has eigenvalues* 3 *and* 0*, and adding* $2\mathbf{1}$ *shifts every eigenvalue by* 2 *without moving any eigenvector:*

$$\operatorname{Spec}(A) = \{5, 2\}, \qquad E_5 = \operatorname{span}\{(1,1,1)\}, \qquad E_2 = \{x : x_1 + x_2 + x_3 = 0\}.$$

Here $\dim E_2 = 2$*. The characteristic polynomial is* $p_A(\lambda) = -(\lambda - 5)(\lambda - 2)^2$*, and* $\operatorname{trace} A = 9 = 5 + 2 + 2$ *while* $\det A = 20 = 5 \cdot 2 \cdot 2$*, each eigenvalue counted twice when it is repeated twice.*

9.3 Over the reals, and over the complexes

Figure 9.3 is worth a look first. The real eigenvalues are the real roots of the characteristic polynomial, and a real polynomial need not have any: for the quarter turn the curve never touches the axis, which is the algebra saying that no direction in the plane survives.



$p(\lambda) = -(\lambda + 1)(\lambda - 1)(\lambda - 2)$

three real eigenvalues

$p(\lambda) = \lambda^2 + 1$

no real eigenvalue at all

Figure 9.3. Real eigenvalues are the real roots of the characteristic polynomial, and a real polynomial need not have any. On the right is the rotation by a quarter turn, whose characteristic polynomial never touches the axis, which is the algebra saying that no direction in the plane survives a rotation. Over $\mathbb{C}$ the picture is different and every polynomial factors completely. The vertical scale has been compressed to fit both curves in one frame.

Theorem 9.9 (Existence over $\mathbb{C}$). *Every operator on a nonzero finite-dimensional complex vector space has an eigenvalue. Over $\mathbb{R}$ this can fail.*

Proof. By Proposition 9.5, p_T is a polynomial of degree $n \geq 1$; by the fundamental theorem of algebra it has a root $\lambda \in \mathbb{C}$, and by Proposition 9.3 that root is an eigenvalue. The failure over $\mathbb{R}$ is Example 9.2(iv): the rotation matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has $p(\lambda) = \lambda^2 + 1$, with no real root. □

Example 9.10 (The rotation, over $\mathbb{C}$). *Regard $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ as acting on $\mathbb{C}^2$. Its eigenvalues are $\pm i$. For $\lambda = i$ we need $(R - i\mathbf{1})v = \mathbf{0}$, that is $-iv_1 - v_2 = 0$, so $v = (1, -i)$ works: indeed $Rv = (i, 1) = i\,(1, -i)$. There is no contradiction with the geometry. Over $\mathbb{R}$ the rotation preserves no line, and over $\mathbb{C}$ what it preserves is not a line in the visible plane.*

Definition 9.11 (Algebraic multiplicity). *Over $\mathbb{C}$, write $p_A(\lambda) = (-1)^n(\lambda - \lambda_1)^{m_1} \cdots (\lambda - \lambda_k)^{m_k}$ with the λ_i distinct. The exponent m_i is the* algebraic multiplicity *of λ_i, and $m_1 + \cdots + m_k = n$.*

Theorem 9.12 (Trace and determinant are the sum and the product). *Let $A \in M_n(\mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ listed with algebraic multiplicity. Then*

$$\operatorname{trace} A = \lambda_1 + \cdots + \lambda_n, \qquad \det A = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Proof. Write $p_A(\lambda) = (-1)^n \prod_i (\lambda - \lambda_i)$. Expanding the product, the coefficient of λ^{n-1} is $-\sum_i \lambda_i$, so the coefficient of λ^{n-1} in p_A is $(-1)^{n+1} \sum_i \lambda_i$. Comparing with Proposition 9.5, which gives $(-1)^{n-1} \operatorname{trace} A$ for the same coefficient, and noting $(-1)^{n+1} = (-1)^{n-1}$, we get $\operatorname{trace} A = \sum_i \lambda_i$. Setting $\lambda = 0$ gives $\det A = p_A(0) = (-1)^n \prod_i (-\lambda_i) = \prod_i \lambda_i$. □

Key Idea

This is why the trace and the determinant were invariant under change of basis in Chapter 7 and Chapter 8: they are the two simplest symmetric functions of the eigenvalues, and the eigenvalues themselves are the real invariants. It also gives a cheap sanity check on any eigenvalue computation, which costs two additions and is worth performing every time.

9.4 How many eigenvectors can there be?

Theorem 9.13 (Distinct eigenvalues give independent eigenvectors)**.** *Let $v_1, \dots, v_k$ be eigenvectors of T for* distinct *eigenvalues $\lambda_1, \dots, \lambda_k$. Then $v_1, \dots, v_k$ are linearly independent.*

Proof. Suppose not. Among all relations $\sum_i c_i v_i = \mathbf{0}$ with not every c_i zero, choose one with the fewest nonzero coefficients, and discard the zero terms, so that every remaining c_i is nonzero. Since each $v_i \neq \mathbf{0}$, at least two terms remain; relabel so they are $i = 1, \dots, r$ with $r \geq 2$.

Applying T to $\sum_{i \leq r} c_i v_i = \mathbf{0}$ gives $\sum_{i \leq r} c_i \lambda_i v_i = \mathbf{0}$. Subtracting λ_1 times the original relation kills the first term:

$$\sum_{i=2}^{r} c_i (\lambda_i - \lambda_1)\, v_i = \mathbf{0}.$$

Every coefficient here is nonzero, because $c_i \neq 0$ and the eigenvalues are distinct, so this is a shorter relation of the same kind. That contradicts the choice of the original one. □

Corollary 9.14. *An operator on an n-dimensional space has at most n distinct eigenvalues. If it has exactly n, then any choice of one eigenvector per eigenvalue is a basis of V.*

Proof. More than n distinct eigenvalues would give more than n independent vectors, contradicting Theorem 6.14(a). With exactly n, the n independent eigenvectors form a basis by Theorem 6.14(c). □

Lemma 9.15 (Block triangular determinant)**.** *If $M = \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$ with A of size $g \times g$ and D of size $(n-g) \times (n-g)$, then* $\det M = \det A \cdot \det D$.

Proof. In the Leibniz formula for M, a permutation σ contributes a nonzero term only if it never picks an entry from the zero block, that is, only if $\sigma(i) \leq g$ for every $i \leq g$. Being a bijection, such a σ permutes $\{1, \dots, g\}$ and $\{g+1, \dots, n\}$ separately, say as α and β, with $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$ since the inversions of σ are those of α together with those of β. Summing over all such pairs factors the sum into the two Leibniz formulas. □

Theorem 9.16 (Geometric multiplicity never exceeds algebraic)**.** *For every eigenvalue λ_0 of an operator T on an n-dimensional space,*

$$1 \;\leq\; \dim E_{\lambda_0} \;\leq\; \textit{the algebraic multiplicity of } \lambda_0.$$

Proof. The lower bound is the definition, since an eigenvalue has a nonzero eigenvector. For the upper bound, put $g = \dim E_{\lambda_0}$, choose a basis $v_1, \dots, v_g$ of E_{λ_0} and extend it to a basis B of V by Theorem 6.14(e). Since $Tv_j = \lambda_0 v_j$ for $j \le g$, the first g columns of $[T]_B$ are $\lambda_0 e_1, \dots, \lambda_0 e_g$, so

$$[T]_B = \begin{pmatrix} \lambda_0 \mathbf{1}_g & B' \\ 0 & C \end{pmatrix}$$

for some blocks B' and C. By Lemma 9.15,

$$p_T(\lambda) = \det\big([T]_B - \lambda \mathbf{1}\big) = (\lambda_0 - \lambda)^g \ \det(C - \lambda \mathbf{1}_{n-g}),$$

so $(\lambda - \lambda_0)^g$ divides p_T up to sign, and the algebraic multiplicity is at least g. □

Example 9.17 (The gap is real). *For the shear $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ we have $p_A(\lambda) = (1-\lambda)^2$, so $\lambda = 1$ has algebraic multiplicity 2. But $A - \mathbf{1} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has a one-dimensional null space, so $\dim E_1 = 1$. Geometrically: the shear leaves the horizontal axis pointwise fixed and tilts everything else, so there is only one invariant direction even though the eigenvalue is doubled. Contrast Example 9.8, where the repeated eigenvalue 2 had a full two-dimensional eigenspace. Which of the two happens is exactly the question this chapter leaves for the next one, on diagonalisation.*

Figure 9.4 shows the shear at work: the horizontal axis stays exactly where it is, and every other point slides sideways by an amount equal to its height, so no second direction can survive.

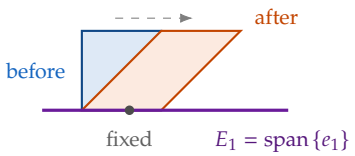


Figure 9.4. Example 9.17. The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ carries the blue square to the orange parallelogram. Every point of the horizontal axis stays exactly where it is, and every point off it slides right by an amount equal to its height. So the eigenvalue 1 is doubled in the characteristic polynomial while its eigenspace is only the one line, which is the inequality of Theorem 9.16 being strict.

9.5 Eigenvalues of related matrices

Proposition 9.18. *Let $Av = \lambda v$ with $v \neq \mathbf{0}$.*

(a) $A^k v = \lambda^k v$ for every $k \geq 1$, and more generally $q(A)v = q(\lambda)v$ for every polynomial q.

(b) If A is invertible then $\lambda \neq 0$ and $A^{-1}v = \lambda^{-1}v$.

(c) An upper or lower triangular matrix has its diagonal entries as eigenvalues, with algebraic multiplicities given by how often each appears.

(d) A and A^t have the same characteristic polynomial, hence the same eigenvalues.

Proof. (a) Induction: $A^k v = A(A^{k-1}v) = A(\lambda^{k-1}v) = \lambda^k v$; then take linear combinations. (b) If $\lambda = 0$ then $v \in N(A)$, contradicting invertibility; applying A^{-1} to $Av = \lambda v$ and dividing by λ gives the claim. (c) For triangular A, $A - \lambda\mathbf{1}$ is triangular with diagonal $a_{ii} - \lambda$, so $p_A(\lambda) = \prod_i (a_{ii} - \lambda)$ by Lemma 8.4. (d) $\det(A^t - \lambda\mathbf{1}) = \det\big((A - \lambda\mathbf{1})^t\big) = \det(A - \lambda\mathbf{1})$ by Proposition 8.13(a). □

Caution

In part (d), only the eigen*values* transfer. The eigenvectors of A^t are in general completely different from those of A; for $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ the eigenvector for $\lambda = 1$ is $(1, 0)$, while for A^t it is $(1, -1)$. The eigenvectors of A^t are sometimes called the left eigenvectors of A, since $v^t A = \lambda v^t$, and keeping the two families apart matters as soon as A is not symmetric.

9.6 How eigenvalues are actually computed

Caution

Do not compute eigenvalues by forming p_A and finding its roots. There are two independent reasons, and both are serious.

First, there is no formula. By the Abel–Ruffini theorem no general polynomial of degree 5 or more can be solved in radicals, and since every monic polynomial is the characteristic polynomial of its companion matrix, any method that produced eigenvalues in finitely many arithmetic operations and roots would solve the quintic. So every eigenvalue algorithm is necessarily iterative. That is a theorem about mathematics, not a gap in anyone's cleverness.

Second, the roots of a polynomial are a violently ill-conditioned function of its

coefficients. Wilkinson's example is $\prod_{k=1}^{20}(x-k)$, whose roots are as well separated as one could wish; perturbing the coefficient of x^{19} by about 2^{-23} moves some roots by more than 2, and sends others off into the complex plane. Passing through the coefficients therefore destroys accuracy that the original matrix had.

Insight

What is used instead works on the matrix directly. The *QR algorithm*, found by Francis and independently by Kublanovskaya around 1961, repeatedly factors and remultiplies until the matrix converges to triangular form, at which point Proposition 9.18(c) reads the eigenvalues off the diagonal; it costs $O(n^3)$ and is on every list of the most important algorithms of the twentieth century. For a single dominant eigenvector there is the older *power method*, which simply applies A over and over to a starting vector and normalises, relying on part (a) above to make the largest λ^k swamp the rest.

Both show up far outside numerical analysis. PageRank is the power method applied to a stochastic matrix of web links. Principal component analysis is the eigenvectors of a covariance matrix, ordered by eigenvalue. Spectral clustering partitions a graph using eigenvectors of its Laplacian. In each case the eigenvector is doing the same job it did for Euler: finding the axes along which the object does something simple.

Open Problem 9.19 (Which lists of numbers are spectra of nonnegative matrices?)**.** *Given a list $\lambda_1, \dots, \lambda_n$ of complex numbers, closed under conjugation, decide whether some $n \times n$ matrix with all entries ≥ 0 has exactly that list as its spectrum. This is the* nonnegative inverse eigenvalue problem*, and* it is unsolved for every $n \geq 5$.

What is known. *Necessary conditions are easy to write down and none of them is sufficient. The Perron root must be present:* $\max_i |\lambda_i|$ *must itself be one of the λ_i, by Chapter 21. The power sums $s_k = \sum_i \lambda_i^k$ are traces of A^k and so must be nonnegative, and they must satisfy $s_k^m \leq n^{m-1} s_{km}$, a condition of Loewy and London. The problem is completely solved for $n \leq 4$, and restricting to lists of* real *numbers buys nothing extra: that case was settled up to $n = 4$ by Loewy and London and is open at $n = 5$ as well. So both versions stop at the same place. What has been settled at $n = 5$ is the trace zero case, by Laffey and Meehan, and that took a substantial argument [215]. There is not even a conjectured answer.*

Where the difficulty sits. *Nonnegativity is a condition on entries, and the spectrum is a condition on a characteristic polynomial. The map between them is the elementary*

symmetric functions of the eigenvalues, which must equal signed sums of principal minors, and the sign constraints do not translate into anything convex. The set of achievable spectra is closed under multiplication by positive scalars and under union in a certain sense, so it is a cone, but not a convex one, and its boundary is where all the difficulty lives. Where a reader might start. *Settle $n = 2$ completely: for which pairs (λ_1, λ_2) does a nonnegative 2×2 matrix exist? The answer is a short computation with the trace and determinant, and it already shows the Perron condition doing work. Then try $n = 3$ with one real and two complex eigenvalues and see how quickly the case analysis grows. Chapter 21 supplies the Perron and Frobenius theory this rests on.*

Exercises

Exercise 9.20. *Find the eigenvalues and a basis of each eigenspace.*

(a) $\left(\begin{smallmatrix} 5 & 2 \\ 2 & 5 \end{smallmatrix}\right)$

(b) $\left(\begin{smallmatrix} 4 & 1 \\ 0 & 4 \end{smallmatrix}\right)$

(c) $\left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 5 & -1 \end{smallmatrix}\right)$

(d) $\left(\begin{smallmatrix} 0 & 1 \\ -2 & -3 \end{smallmatrix}\right)$

In each case check your answer against Theorem 9.12.

Exercise 9.21. *For each matrix in Exercise 9.20, state the algebraic and geometric multiplicity of every eigenvalue, and say where Theorem 9.16 is an equality and where it is strict.*

Exercise 9.22. *Prove that A and $A + c\mathbf{1}$ have the same eigenvectors, with eigenvalues shifted by c. Then explain the trick used in Example 9.8 in those terms, and use it to find the eigenvalues of the $n \times n$ matrix with a on the diagonal and b everywhere else.*

Exercise 9.23. *Let P satisfy $P^2 = P$, an operator of this kind being called a projection. Show that every eigenvalue of P is 0 or 1. Then show that $V = \ker P \oplus \operatorname{Im} P$ in the sense that every v splits uniquely as $(v - Pv) + Pv$.*

Exercise 9.24. *Let N satisfy $N^k = 0$ for some k. Prove that the only eigenvalue of N is 0, and deduce that $p_N(\lambda) = (-1)^n \lambda^n$ over $\mathbb{C}$. Give a 3×3 example whose eigenspace for 0 has dimension exactly 1.*

Exercise 9.25. *Find the complex eigenvalues and eigenvectors of the rotation matrix $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, and check that their sum is the trace and their product the determinant. For which θ are the eigenvalues real, and what is the map then?*

Exercise 9.26. *Suppose $AB = BA$ and v is an eigenvector of A for an eigenvalue λ whose eigenspace E_λ is one-dimensional. Show that v is also an eigenvector of B.* Hint: show first that B maps E_λ into itself.

Exercise 9.27. *Let $q(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ and let C be the $n \times n$ matrix with ones on the subdiagonal, $-c_0, \dots, -c_{n-1}$ in the last column, and zeros elsewhere. Verify for $n = 3$ that $p_C(\lambda) = (-1)^3 q(\lambda)$. Explain why this makes the claim about the quintic in Section 9.6 correct.*

Exercise 9.28. *Take $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $x_0 = (1, 0)$. Compute $x_{k+1} = Ax_k / \|Ax_k\|$ for $k = 0, 1, 2, 3$ and watch the sequence approach an eigenvector. Which one, and why that one rather than the other?*

References and further reading

The route by which eigenvalues entered mathematics, through rigid-body mechanics and the secular equations of celestial mechanics, is worth reading about; it explains the otherwise puzzling vocabulary. Axler [37] proves Theorem 9.9 without determinants at all, by observing that $v, Tv, \dots, T^n v$ must be dependent and factoring the resulting polynomial, and that argument is more illuminating than ours even though ours is shorter given what Chapter 8 has already built. Hefferon [307] and Austin [35] both supply more computational practice than we have room for, and Austin is good on the geometric reading of Figure 9.2. Banica [51], our baseline, moves quickly from here to the spectral theorems, which are the point of the whole development. For the computational side, and in particular for the QR algorithm and for why the characteristic polynomial must never be formed, Trefethen and Bau [607] devote several lectures to exactly this material; their discussion of Wilkinson's polynomial is the standard reference for how bad the conditioning can be.